Assignment Markup Language™

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Assignment Markup Language™ (AML) is a scripting language that enables instructors, instructional designers, and other assignment creators to embed assignment-administration directives with assignment content in the document file of a word processor. The combination can be read by people using any word processor and executed by an assignment management application (AMA). Using AML, assignment creators develop assignments independent of the AMA that delivers the assignment to students, enabling content editing and licensing independent of assignment administration.

AML is compatible with hypertext markup language (HTML), but whereas HTML commands software to present items in the same way to all parties, AML directs software to present different items differently depending on the party targeted. The targeted parties include the *respondent* who submits the assignment, the *administrator* who oversees the assignment, the *jury member* who rates the submission, and the *grader* who scores selected items of the submission. Often, the respondent is the student, the administrator is the instructor, the jury consists of members of the student's class who are not respondents of the assignment, and the grader is the instructor's teaching assistant.

AML directives fall into four categories: open-ended, closed-ended, screening, and miscellaneous. An open-ended directive does not impose an item-specific limit on the number of characters that can be entered in response to the item. A closed-ended directive imposes an item-specific limit. A screening directive hides a part of the assignment from the view of a targeted party at a particular time. Miscellaneous directives include a directive for the administrator to add or to subtract credit, and an implied directive to present the respondent with a blank page for unrestricted typing with a default format.

Directives are constructed by combining ten keyboard characters: vertical pipe (|), tilde (~), greater than (>), less than (<), underscore (_), hash (#), hyphen (-), plus (+), equal (=), and comma (,). Of these, the vertical-pipe and tilde characters are reserved: The vertical-pipe character should appear in an AML document only in conjunction with an AML directive, and the AMA must neither accept the respondent's entry of a vertical-pipe character on an open-ended item nor accept it or the tilde on a closed-ended item. The directives are listed in Table 1 through Table 3.

To create an assignment with AML, the instructor creates a master form embedded with directives and sets aside spaces in the document for responses. Although the AMA could comply with the directives of the master form by producing a product that appears very different from the master form, AML is designed to enable the AMA to produce that product by copying and editing the master form itself, thereby requiring much less application developmental work. The AMA

described herein uses the master form in this way. Relationships are illustrated by two flowcharts: Figure 1 and Figure 2.

As Figure 1 shows, the master form is input to the AMA that produces a fillable form for the respondent and a copy of the master form for the grader. The respondent's filled form and the grader's edited copy of the master form are returned to the AMA, where data are extracted and processed.

As Figure 2 shows, the AMA computes the keyed score of each closed-ended item. For the open-ended items, the AMA applies the administrator's weight to balance between an item-scoring formula accounting for both the grader's score and self-score of each open-ended item, and holistic jury ratings of all the items. Then the AMA sums the scores to arrive at the grade for the respondent's submission.

Table 1
Open-Ended Directives

*Directive**	*Description*
>…	1. Immutable open-ended. If appearing as the first character of a paragraph or leftmost row of a table, the respondent can type freely after the greater-than character (>) for the rest of the paragraph or the remaining cells of the row.
|…	2. Mutable open-ended. Like the immutable open-ended directive, except that the vertical-pipe character (|) can be deleted. Its deletion causes the editable paragraph to either be appended to the immediately preceding editable paragraph or become un-editable if not so preceded. Within a table, its deletion causes all the cells of its row to be un-editable.
<>|__|*L*|κ|β|*e*|>…	3. Scorable, immutable open-ended. The respondent can enter within the spaces between the first two vertical-pipe characters a numerical self-score that does not exceed *L*. If the respondent's self-score exceeds the grader's score, the item score is reduced by the excess of the respondent's self-score over the grader's score multiplied by the self-enhancement penalty, κ. Thus, when Self Score > Grader's Score, Basic Score = Grader's Score – κ (Self Score – Grader's Score); otherwise, Basic Score = Grader's Score. Should the difference between the grader score and self-score be within the tolerance of ±*e*, the item score is increased by the point value of the bonus, β. Thus, when the | Self Score – Grader's Score | ≤ *e*, Item Score = Basic Score + β, otherwise Item Score = Basic Score. Default values for items omitted are *L* = 5, κ = 1, β = .5, and *e* = 0. The optional less-than character on the left end directs the AMA to insert the Item Score into the submission when the item is graded.

*The letters *L*, κ, β, and *e* stand for numbers. Ellipsis (…) stands for unlimited text. Underscores (__) represent spaces allowed for the respondent's entry.

Table 2
Closed-Ended Directive

*Directive**	*Description*
<>|_*aa*_|*x,r*|*abc*|*ab*|*ac=y*|…|<>	The respondent's entry is limited in characters to the number of spaces between the first two vertical-pipe characters (|). Credit applies if the entry matches, case-insensitive, the *aa* key. The item's credit value is *x* and the number of retries[1] allowed is *r*. The respondent can type only the *abc* characters, in any combination and any order. Exceptions to the key are *ab*, *ac*, and possibly fewer or more. Default values for items omitted are *x* = 1 and *r* = 5. If the key is specified, credit applies to any entry that matched either the key or one of its exceptions. If the key is not specified, credit applies to any entry that does not match an exception. Credit can be specified in conjunction with an exception by appending the equal (=) character to the exception followed immediately by the credit value of the exception. Credit specification overrides default treatment. Thus, when the item's value is *x*, and the exception is specified as *ac=y*, credit received for responding with *ac* is *y*. The tilde (~), signifying the absence of an entry, can be specified as the key or an exception, but credit for the absent-entry item applies only when the assignment is modified. Therefore, the assignment with a credit-for-no-entry item also should include at least one credit-for-entry item. The optional less-than character on the left end directs the AMA to insert the item's score and *aa* key when the item is graded. The optional greater-than character immediately following the less-than character on the right end directs the AMA to insert the item's credit value and number of retries allowed on the respondent's fillable form. Thus, if *x* =1 and *r* = 2, the AMA's insertion will appear as <1,2>. Hint: For better clarity, the same information can be entered manually by immediately following the less-than character with a blank character, like this, < 1 point, 2 retries>.

* The letters *a*, *b*, and *c* stand for strings of characters. The letters *x*, *r*, and *y* stand for numbers. Ellipsis (…) stands for unlimited text. Underscores (_) represent spaces allowed for the respondent's entry.
[1]Retries apply only if the AMA allows the respondent who submitted an incorrect answer to try again.

Table 3
Screening and Miscellaneous Directives

*Directive**	*Description*
--|…|--	5. Comment screening. The in-between text (…) is viewable only by administrators.
++|…|++	6. Explanation screening. The in-between text (…) is viewable by respondents only after the assignment's graded date.
-+|…|+-	7. Flash screening. The in-between text (…) is viewable by respondents only when they are working on the assignment.
+-|…|-+	8. Not-for-judge screening. The in-between text (…) is not viewable by either jury or grader when submissions are being judged.
#|…|#	9. First to penultimate reorderable item of a section.
#|…|##	10. Last reorderable item of a section.
~>*x*…	11. Free credit. The submission will be additionally credited (or debited) with the *x* points, explanation (…) allowed.
[Document of 100 or fewer characters]	12. Blank page. The default format is that of the characters, which does not appear. The respondent can type freely on the blank page.

* The letters *a*, *b*, and *c* stand for strings of characters. The letter *x* stands for a number. Ellipsis (…) stands for unlimited text.

Figure 1
Flow Chart Showing How Forms are Managed

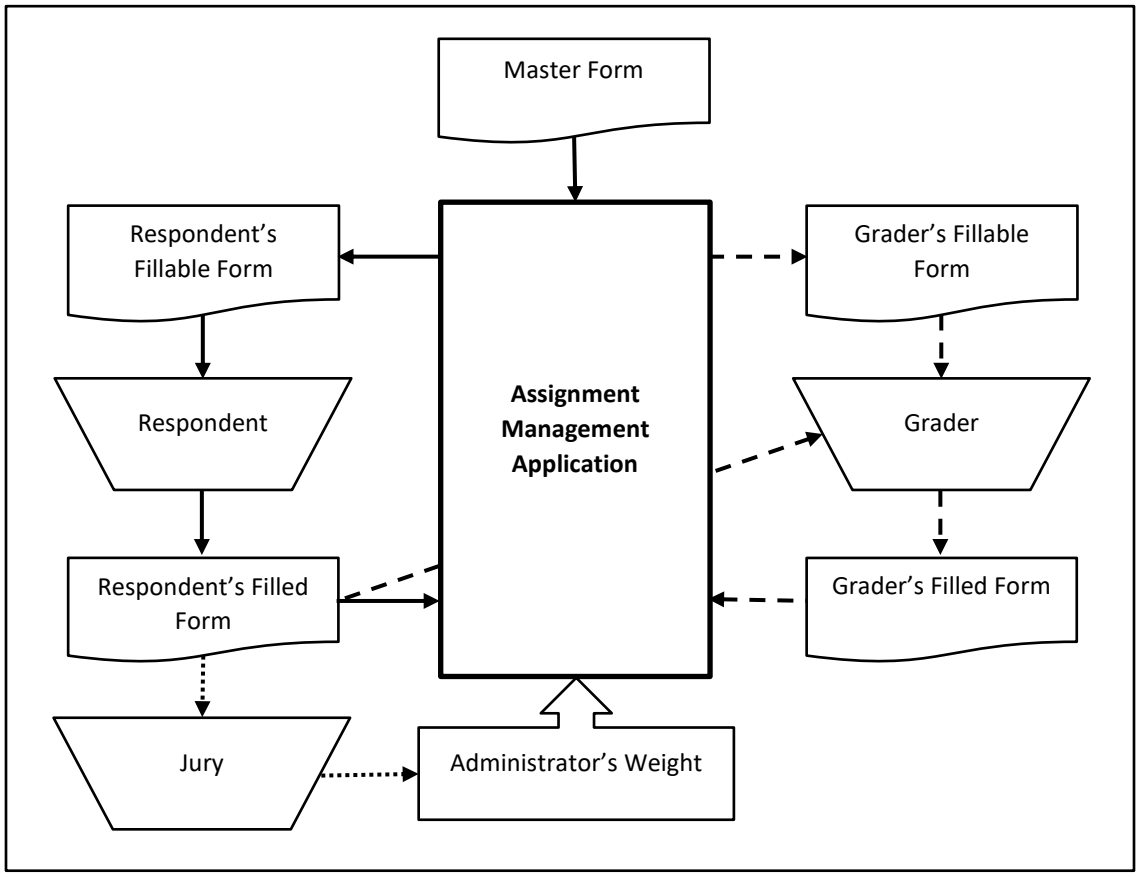


Figure 2
Flow Chart Showing How Grades Are Derived

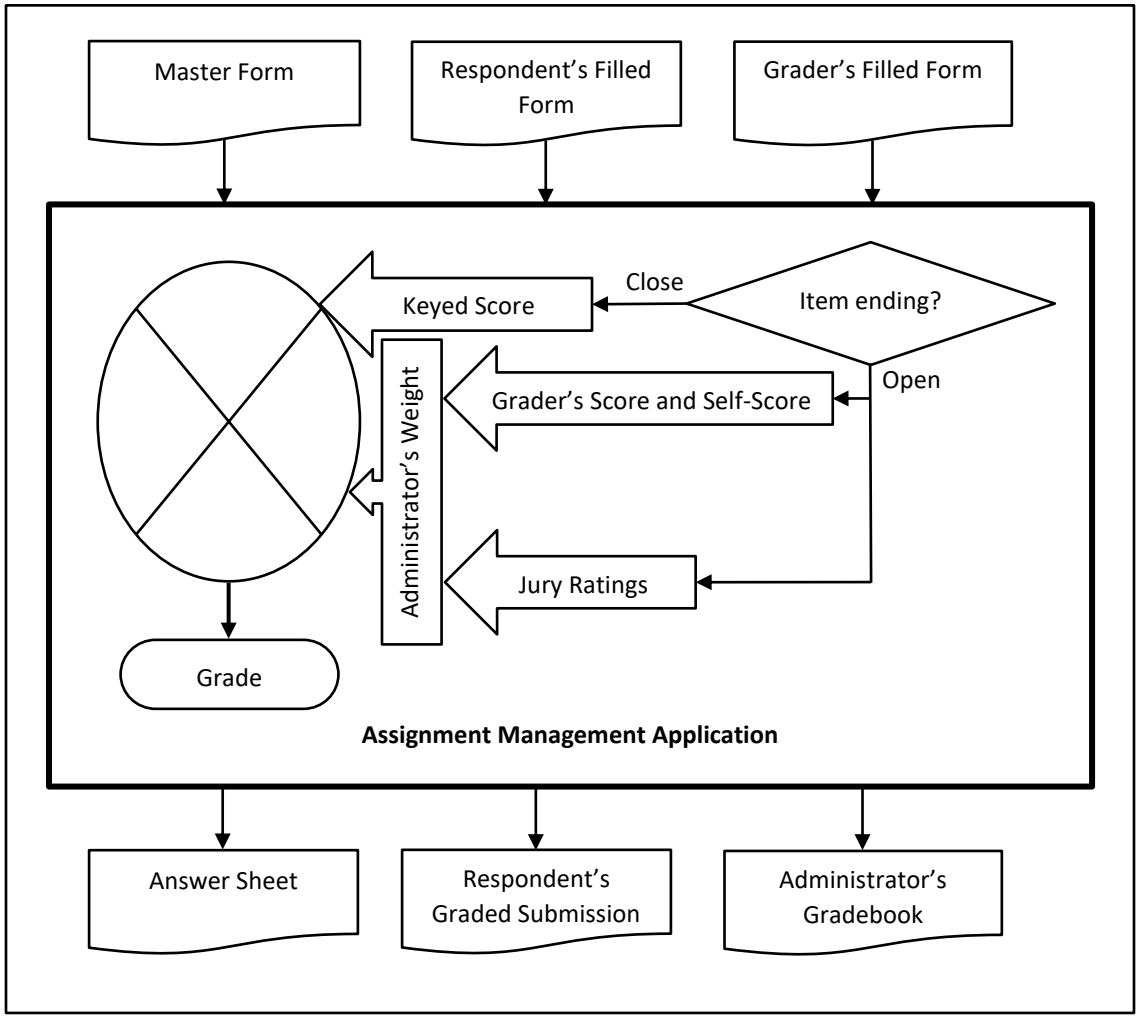


Figure 3 shows the master form of a quiz on astronomy that contains AML directives with content as appearing on a Microsoft Word document. The quiz contains the following items:

Item 1. Self-scored, open-ended essay. The correct answer is "*A virtually uncountable ….*" The highest acceptable self-score entry is 5 points.
Item 2. Fill in the blanks. The key is *planet*. The score for entering the keyed response, not specified, is the AMA's default value, perhaps one point.
Item 3. True or false. The key is *F*. The respondent is to be allowed to enter only an upper or lower case of *t* or *f*. The score for entering the keyed response, again not specified, is the AMA's default value.
Item 4. Multiple choices. The key is *C*. The allowed single-character response is *a* or *b* or *c* or *d*. The score for entering the keyed response is specified: 2 points.
Item 5. Multiple answers. The key is *Y* for the first two answers and *N* for the remaining two answers. The score for entering the keyed response, .50, and the allowed single-character entry, *y* and *n*, are the same for all four answers.

Figure 4 shows the fillable form that the AMA generates for the respondent. It differs from the master form in the following ways.

Item 1. Self-scored, open-ended essay. The correct answer and highest acceptable self-score entry are omitted. The AMA assures that the respondent cannot enter a self-score higher than the specified 5 point.
Item 2. Fill in the blanks. The key is omitted.
Item 3. True or false. The key and allowable characters are omitted. The AMA assures that the respondent can enter only one character, *t* or *f*.
Item 4. Multiple choices. The key, point value, and allowable characters are omitted.
Item 5. Multiple answers. The key, point value, and allowable characters are omitted from every answer.

Figure 5 shows the filled-form that a respondent might submit. Figure 6 shows the jury member's and grader's view of the submission, wherein the respondent's self-score is replaced by a period (.) and the items between the opening and closing not-for-judging flags are removed, to forestall bias.

AML is a scripting language for assignments. AML enables assignments to be created, edited, and licensed independent of the AMA that administers assignments.

Figure 3
Master Form of Assignment

Quiz

1. Define the word *galaxy* as used in astronomy.

>|5|> ++| A virtually uncountable collection of stars held together by gravity. |++

+-|
2. A moon is a natural satellite that revolves around a >|____planet_____|<.

3. True or false: Earth is closer to the Sun than Mars. >|F||tf|<

4. Which one of the following planets is farthest from the Sun? Type the letter of your selection into the underscored space below the list of choices. (2 points)

a. Uranus
b. Jupiter
c. Neptune
d. Saturn

>|C|2|abcd|<

5. Which one of the following are the two smallest planets by size? Enter *y* into the underscored space to the left of each of the two smallest planets and Enter *n* into the corresponding space of each of the two other planets. (2 points)

>|Y|.50|yn|< a. Mercury
>|Y|.50|yn|< b. Neptune
>|N|.50|yn|< c. Jupiter
>|N|.50|yn|< d. Earth
|-+

Figure 4
Fillable Form of Assignment

Quiz

1. Define the word *galaxy* as used in astronomy.

>|_|>

+-|
2. A moon is a natural satellite that revolves around a >|_____|<.

3. True or false: Earth is closer to the Sun than Mars. >|_|<

4. Which one of the following planets is farthest from the Sun? Type the letter of your selection into the underscored space below the list of choices. (2 points)

a. Uranus
b. Jupiter
c. Neptune
d. Saturn

>|_|<

5. Which one of the following are the two smallest planets by size? Enter *y* into the underscored space to the left of each of the two smallest planets and Enter *n* into the corresponding space of each of the two other planets. (2 points)

>|_|< a. Mercury
>|_|< b. Neptune
>|_|< c. Jupiter
>|_|< d. Earth
|-+

Figure 5
A Respondent's Submission of the Assignment

Quiz

1. Define the word *galaxy* as used in astronomy.

>|3|> A bunch of planets around a star.

+-|
2. A moon is a natural satellite that revolves around a >|___ planet_____|<.

3. True or false: Earth is closer to the Sun than Mars. >|F|<

4. Which one of the following planets is farthest from the Sun? Type the letter of your selection into the underscored space below the list of choices. (2 points)

a. Uranus
b. Jupiter
c. Neptune
d. Saturn

>|A|<

5. Which one of the following are the two smallest planets by size? Enter *y* into the underscored space to the left of each of the two smallest planets and Enter *n* into the corresponding space of each of the two other planets. (2 points)

>|Y|< a. Mercury
>|Y|< b. Neptune
>|N|< c. Jupiter
>|N|< d. Earth
|-+

Figure 6
Jury Member's and Grader's View of the Submission

Quiz

1. Define the word *galaxy* as used in astronomy.

>|.|> A bunch of planets around a star.

+-||-+